
Week 4 – Task 11 (SOC)

Ingest Logs into SIEM + Basic Searches

Field	Details
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1. Introduction

The objective of this task was to understand Security Information and Event Management (SIEM), ingest Windows security logs into Wazuh, verify that the Windows Server 2022 endpoint was communicating with the Wazuh server, identify failed login events, perform basic searches, filter events, and analyze the resulting security events.

The practical environment for this task consisted of the following components:

- Wazuh SIEM running on Kali Linux
- Windows Server 2022 as the monitored endpoint
- Wazuh Agent installed on Windows Server 2022
- Wazuh Dashboard used for event monitoring and searching

2. Objectives

Objective	Description
SIEM Log Ingestion	Ingest Windows security events into Wazuh
Agent Deployment	Verify the Wazuh Agent on Windows Server 2022
Security Log Collection	Confirm collection of the Windows Security event channel
Event Monitoring	Monitor events generated by the Windows endpoint
Failed Login Detection	Identify failed login events
SIEM Searching	Perform searches using Wazuh query syntax
Event Filtering	Filter events based on rule ID and agent
Time-Based Analysis	Use a time range to review security events

3. Lab Environment

Component	Details
SIEM Platform	Wazuh
SIEM Server	Kali Linux
Wazuh Server IP	10.209.222.27
Monitored Endpoint	Windows Server 2022
Windows Endpoint IP	10.209.222.18
Wazuh Agent	v4.14.5

Component	Details
Agent Name	WIN-BCDCP359AAS
Agent ID	004
Operating System	Microsoft Windows Server 2022 Standard Evaluation
Agent Status	Active

The Windows Server 2022 endpoint was successfully registered with the Wazuh server and appeared as an active agent in the Wazuh Dashboard, confirming that agent-to-manager communication was established correctly.

4. Wazuh Agent Deployment and Verification

The following technical process was used to deploy and verify the Windows Wazuh Agent:

- Windows Server 2022 was configured as the monitored endpoint.
- The Wazuh Agent configuration was checked.
- The Wazuh Manager server address was configured as 10.209.222.27.
- The Wazuh Agent service was restarted after configuration.
- The Wazuh service was verified as running.
- The Wazuh Dashboard was checked to confirm that the endpoint was active.

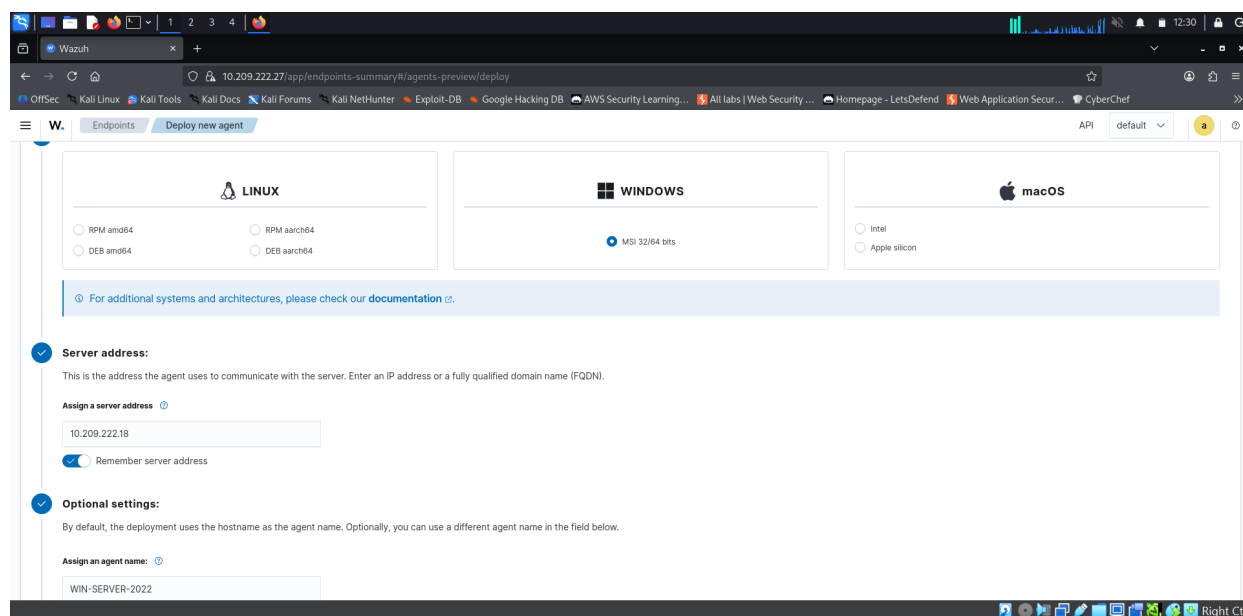


Figure 1: Wazuh Dashboard – Windows Agent Deployment Configuration

The deployment screen shows the Windows agent package selection (MSI 32/64 bits) along with the assigned server address of 10.209.222.27 and the agent name configured as WIN-SERVER-2022, confirming the deployment parameters used to register the endpoint with the Wazuh manager.

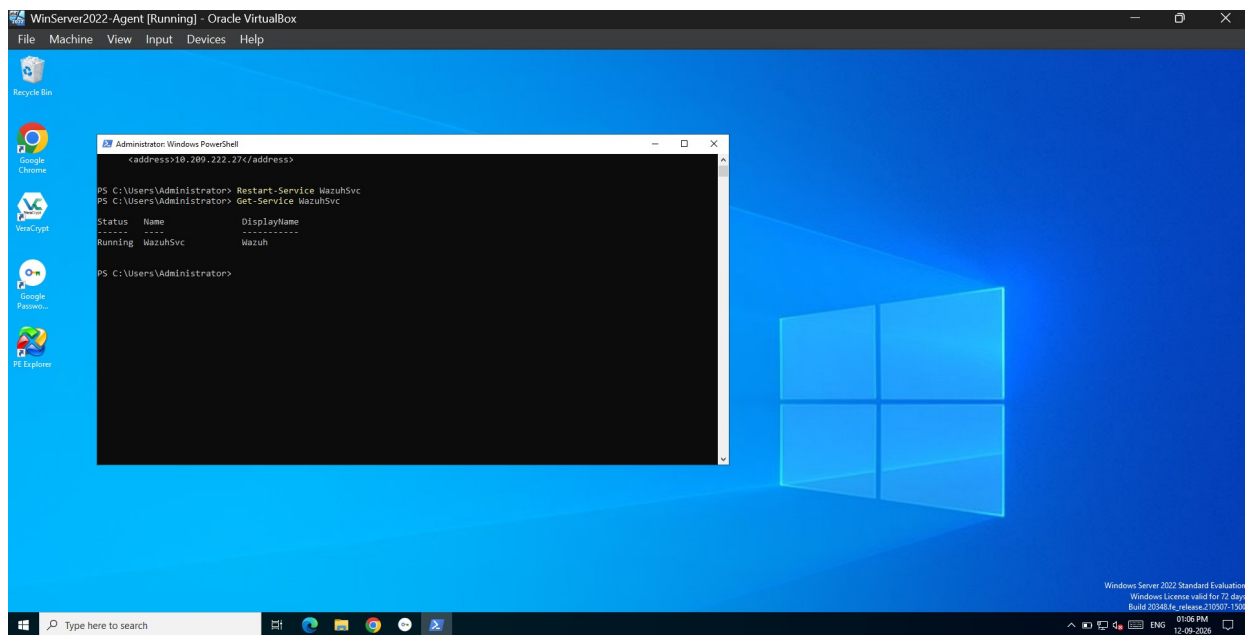


Figure 2: Windows Server 2022 – Wazuh Agent Service Running

This screenshot demonstrates that the WazuhSvc service on the Windows Server 2022 endpoint is in the Running state, verified using the Get-Service WazuhSvc PowerShell command following a service restart.

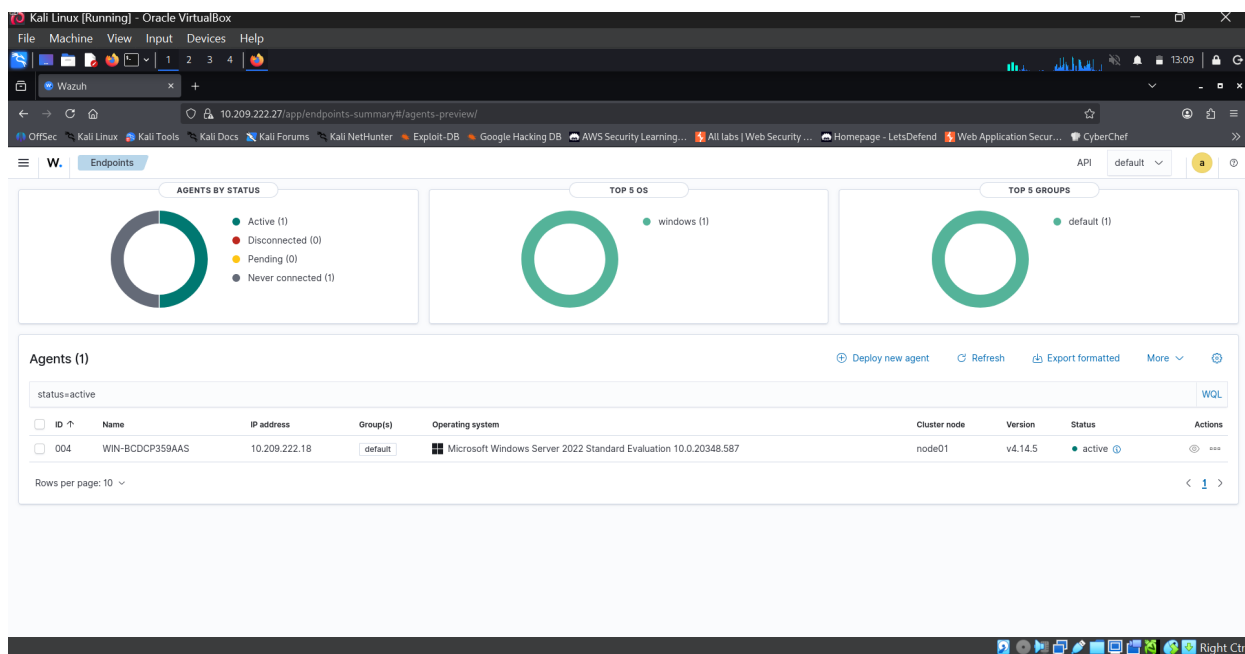


Figure 3: Wazuh Dashboard – Windows Server 2022 Agent Active

The Wazuh Dashboard confirms that the endpoint is registered and communicating with the manager. The Endpoints view shows Agent ID 004, Agent name WIN-BCDCP359AAS, IP address 10.209.222.18, operating system Microsoft Windows Server 2022 Standard Evaluation, Wazuh Agent version v4.14.5, and status active.

5. Windows Security Log Collection

Windows Security events were checked directly on the endpoint using PowerShell to confirm that failed login activity was being generated locally before verifying ingestion into Wazuh. The following command was used:

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} -MaxEvents 10
```

Windows Event ID 4625 represents failed logon activity. This command was used to verify that failed login events were present in the Windows Security log on the monitored endpoint.

The Wazuh Agent configuration file was then checked to confirm that the Security event channel was configured for collection, using the following command:

```
Select-String -Path "C:\Program Files (x86)\ossec-agent\ossec.conf" -Pattern "<location>Security</location>"
```

The output confirmed the presence of the following configuration entry:

```
<location>Security</location>
```

This confirmed that the Wazuh Agent on Windows Server 2022 was configured to monitor and forward the Windows Security event channel to the Wazuh manager.

6. Wazuh Event Ingestion

After restarting the Wazuh Agent service, the Wazuh Dashboard was checked for events originating from the Windows Server 2022 endpoint. The Threat Hunting event view showed a large number of events associated with the agent WIN-BCDCP359AAS.

The dashboard displayed 499 hits when filtering by the following query:

```
agent.name: "WIN-BCDCP359AAS"
```

This confirmed that Wazuh was actively receiving and displaying events forwarded from the Windows endpoint.

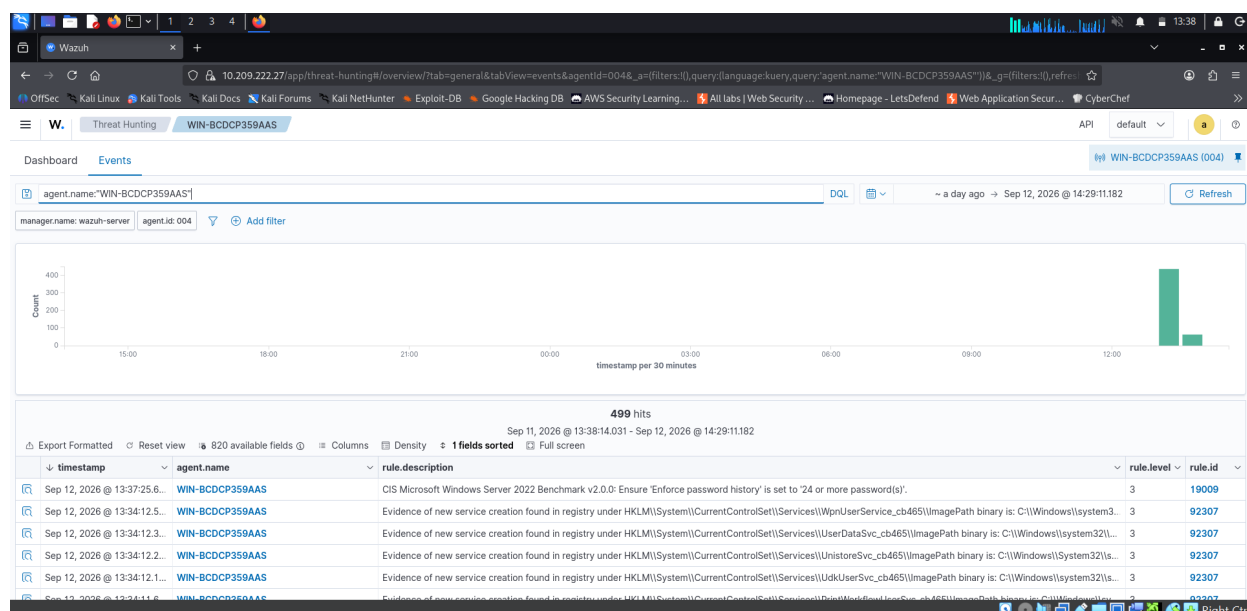


Figure 4: Wazuh Threat Hunting – Events Received from Windows Server 2022

The event view shows the agent name WIN-BCDCP359AAS along with 499 hits over the selected time range, indicating consistent event ingestion including service-creation and CIS benchmark compliance events from the endpoint.

7. Failed Login Event Detection

Failed login events were identified in Wazuh using the following query:

```
rule.id:60122
```

The search returned 6 hits, with the displayed rule description "Logon Failure - Unknown user or bad password". The events were associated with agent WIN-BCDCP359AAS, rule ID 60122, and rule level 5.

These events demonstrate that Wazuh successfully detected failed authentication activity generated on the Windows Server 2022 endpoint.

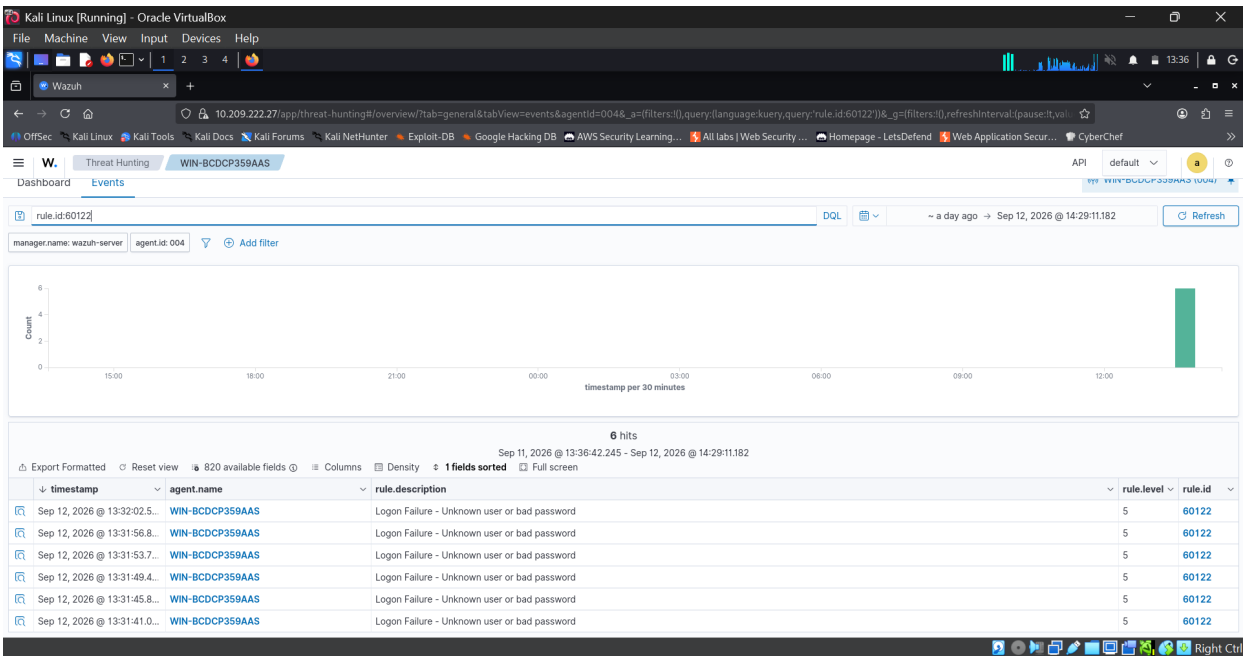


Figure 5: Wazuh – Filtered Failed Login Events Using Rule ID 60122

Key findings from this search:

- 6 failed-login events
- Rule ID 60122
- Rule Level 5
- Logon Failure – Unknown user or bad password
- Windows agent WIN-BCDCP359AAS

8. Basic SIEM Searches

Search Purpose	Wazuh Query	Result / Observation
View all events from Windows	agent.name:"WIN-	499 hits displayed

Search Purpose	Wazuh Query	Result / Observation
agent	BCDCP359AAS"	
Identify failed login alerts	rule.id:60122	6 hits displayed
Identify failed Windows logon events	Windows Event ID 4625	Failed logon activity identified on the Windows endpoint

Filtering by the agent name allowed the analyst to view all events generated by the specific Windows Server 2022 endpoint, providing visibility into the full range of activity forwarded to the SIEM.

Filtering by rule ID 60122 isolated failed authentication alerts, allowing the analyst to focus specifically on logon-failure activity without reviewing unrelated events.

Windows Event ID 4625 was used at the Windows endpoint level to independently verify that failed authentication events were being generated and logged prior to confirming their ingestion into Wazuh.

9. Time-Range Filtering

Time-based filtering is an important technique in SIEM analysis. The Wazuh event interface was used with a defined time range to narrow the investigation to relevant events. Time filtering helps SOC analysts to:

- Reduce irrelevant events
- Investigate activity within a specific period
- Correlate events
- Identify recent security activity
- Improve investigation efficiency

10. TryHackMe – Introduction to SIEM

The mandatory free TryHackMe "Introduction to SIEM" room was completed as part of the learning requirement for this task. The room covered foundational SIEM concepts and features, including:

- SIEM fundamentals
- Log sources and ingestion
- Security event analysis
- Alerting and analysis

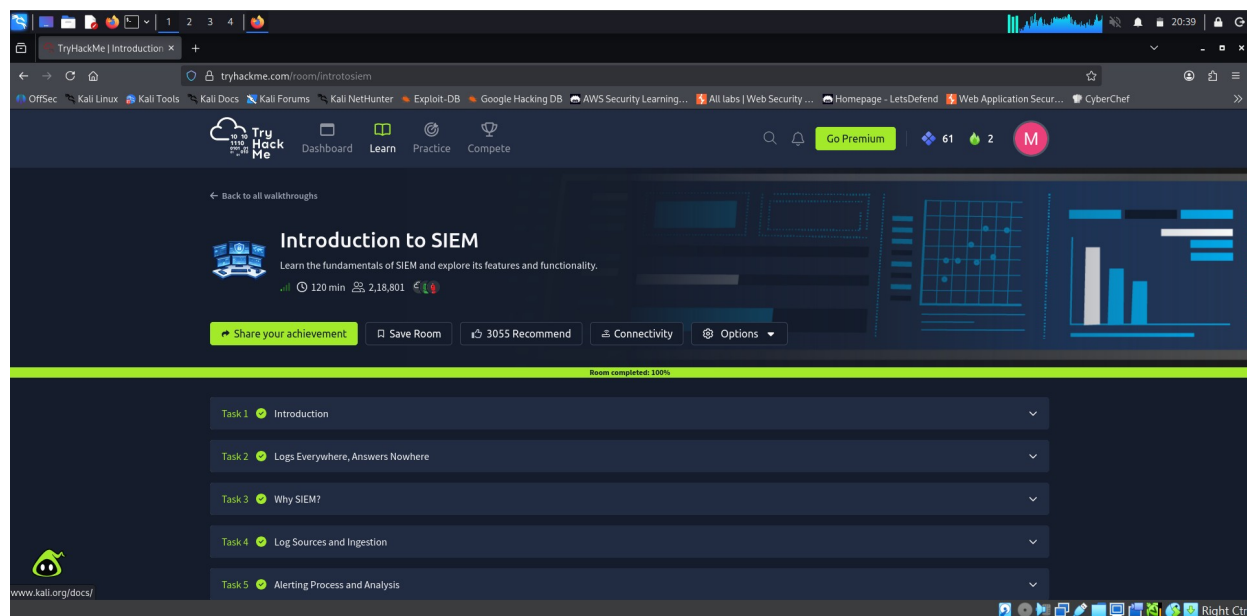


Figure 6: TryHackMe – Introduction to SIEM Room Completed

The screenshot shows the room completed at 100%, providing evidence of completing the required TryHackMe SIEM learning activity.

11. Findings and Observations

1. The Windows Server 2022 Wazuh Agent was successfully connected to the Wazuh server.
2. The agent appeared as Active in the Wazuh Dashboard.
3. The Windows Security event channel was configured for collection.
4. Windows Event ID 4625 failed-login events were present on the endpoint.
5. Wazuh successfully detected failed authentication activity.
6. Wazuh Rule ID 60122 identified "Logon Failure - Unknown user or bad password".
7. Six failed-login alerts were identified through the Rule ID 60122 search.
8. A broader agent-based search returned 499 events, demonstrating active event ingestion from the endpoint.
9. Time-range filtering was used to narrow the SIEM investigation to relevant periods.
10. The TryHackMe Introduction to SIEM room was completed at 100%.

12. Security Analysis

Failed authentication events are relevant security indicators because repeated failed login attempts may indicate:

- Incorrect credentials
- User authentication problems
- Automated authentication attempts
- Possible password guessing or brute-force activity when occurring at higher volume

The six failed-login events observed in this task should not be classified as a brute-force attack based on this evidence alone. They are failed authentication events, and additional context — including frequency, source information, and correlation with other activity — would be required before classifying them as malicious.

A SOC analyst would typically investigate the following factors to reach a conclusion:

- Source IP
- Target username
- Authentication type
- Failure reason
- Number of attempts
- Time pattern
- Successful login events following failures
- Related endpoint activity

13. Skills Demonstrated

Skill	Demonstrated Activity
SIEM	Wazuh deployment and event monitoring
Log Ingestion	Windows Security event collection
Windows Event Analysis	Event ID 4625 investigation
Event Filtering	Wazuh Rule ID 60122 filtering
Threat Hunting	Searching endpoint events
Authentication Monitoring	Failed login detection
PowerShell	Windows event log querying
SOC Analysis	Basic security event investigation
Time-Based Analysis	SIEM time-range filtering
Documentation	Evidence collection and technical reporting

14. Conclusion

This task provided practical experience with SIEM-based log ingestion and basic security event analysis using Wazuh. The Windows Server 2022 endpoint was successfully integrated with Wazuh, Security events were collected, failed authentication events were identified, Wazuh searches were performed, and the mandatory TryHackMe Introduction to SIEM room was completed.

This activity strengthened practical SOC fundamentals, including log monitoring, event filtering, authentication-event analysis, and basic SIEM investigation.

15. Evidence Summary

Figure	Screenshot File	Evidence
Figure 1	01_wazuh_deploy_new_agent.png	Wazuh agent deployment configuration
Figure 2	02_wazuh_agent_service_running.png	Wazuh Agent service running
Figure 3	03_wazuh_agent_active_dashboard.png	Windows Server 2022 agent active in Wazuh
Figure 4	05_wazuh_all_events_from_windows_agent.png	Events received from Windows endpoint
Figure 5	04_wazuh_failed_login_events_rule_60122.png	Six failed-login events detected
Figure 6	06_tryhackme_intro_to_siem_completed.png	TryHackMe Introduction to SIEM completed at 100%